

3.1 Fundamentals of algorithms

3.1.1 Representing algorithms

Content	Additional information
Understand and explain the term algorithm.	An algorithm is a sequence of steps that can be followed to complete a task. Be aware that a computer program is an implementation of an algorithm and that an algorithm is not a computer program.
Understand and explain the term decomposition.	Decomposition means breaking a problem into a number of sub-problems, so that each sub-problem accomplishes an identifiable task, which might itself be further subdivided.
Understand and explain the term abstraction.	Abstraction is the process of removing unnecessary detail from a problem.
Use a systematic approach to problem solving and algorithm creation representing those algorithms using pseudo-code, program code and flowcharts.	Any exam question where students are given pseudo-code will use the AQA standard version. Exam questions will indicate the form of response expected. For example, pseudo-code, program code or a flowchart.
Explain simple algorithms in terms of their inputs, processing and outputs.	Students must be able to identify where inputs, processing and outputs are taking place within an algorithm.
Determine the purpose of simple algorithms.	Students should be able to use trace tables and visual inspection to determine how simple algorithms work and what their purpose is.